

Campus Climate Action

Principles of Software Engineering
Sec 14- Mr. Mohd Noor Hafizee Yahaya



Group members:

1. Gao Simeng
2. Yao Longjun
3. Felia Alendra
4. Hong Seohee

Problem

Climate change has become one of the biggest issue in the world today. Most of human activity produce massive amount of carbon dioxide- which is the main readon for climate change.

It affects everyone and the ecosystem, creating unhelathy environment for all of the living organism.

SDG 13

Therefore we decide to make this app as a response to climate action. We realize that university has significant influence in terms of young people becoming active participants in climate action.



Campus Climate Action - App

Target User: All Taylor's Student

Main feature:

1. Join challenge
2. Redeem points
3. Track CO2 footprint





Product Vision

- Encourage student to commit low-carbon activity
- Empower student to take action on climate change
- Provide platform for university to create sustainable ecosystem

Agile & SCRUM

Agile Methodology:

- iterative & incremental approach to software development
- focuses on flexibility, collaboration, and customer feedback
- work is divided into short cycles (sprints) for continuous improvement
- encourages adaptive planning and quick response to change

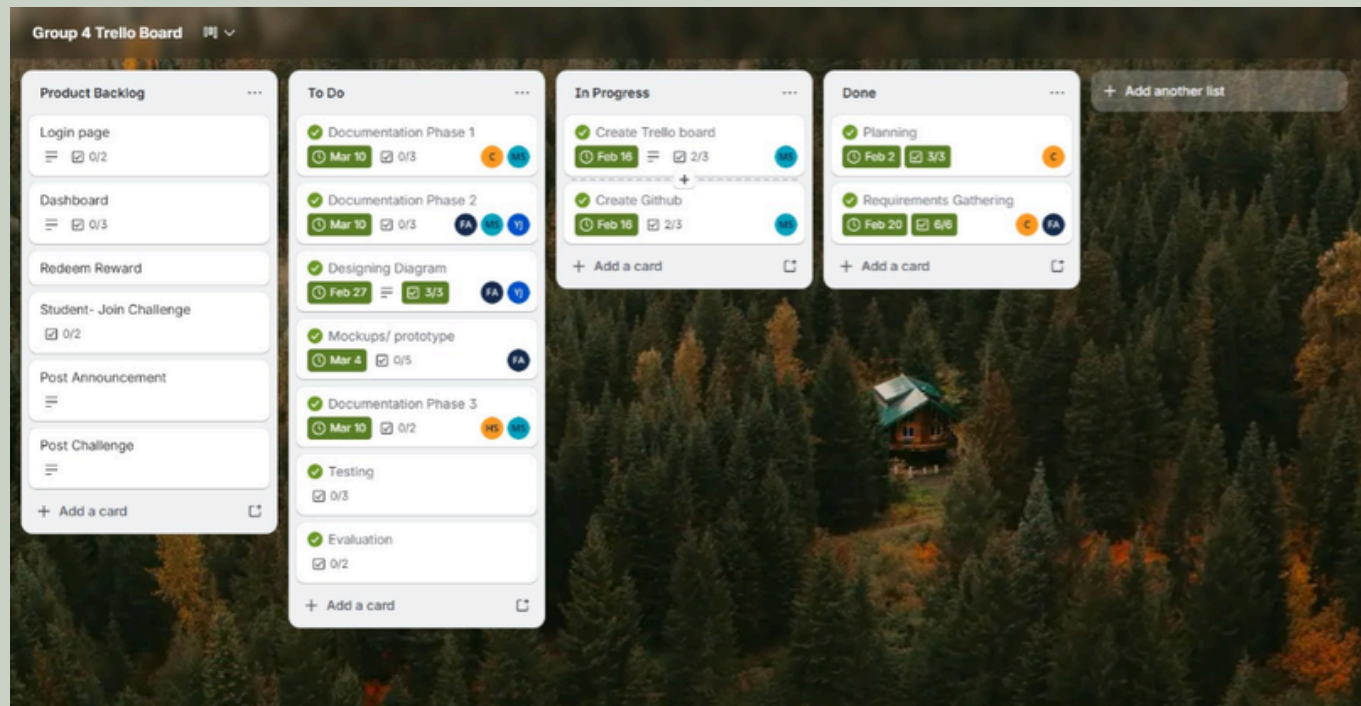
Sprint Planning

- Meeting at the start of each sprint
- Team defines goals, tasks, and priorities for the sprint
- Breaks down user stories into actionable items
- Ensures shared understanding of what will be delivered

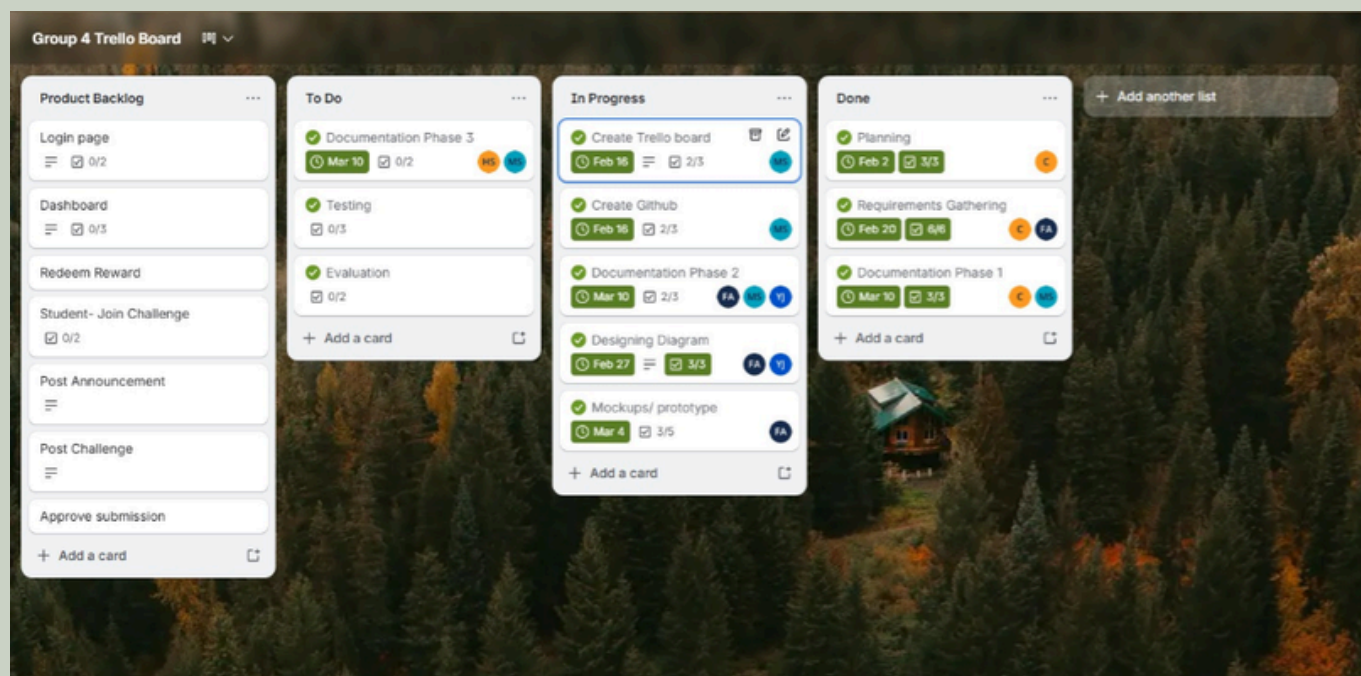
Sprint	Focus	Key activities
sprint 1	Foundation	set up basic app structure, UI design, initial announcement feature
sprint 2	Functionality	Add core features (eg. announcement interaction, reward system), refine usability
sprint 3	Enhancement	Improve accessibility, add profile editing, polish design, testing&feedback



Week 1

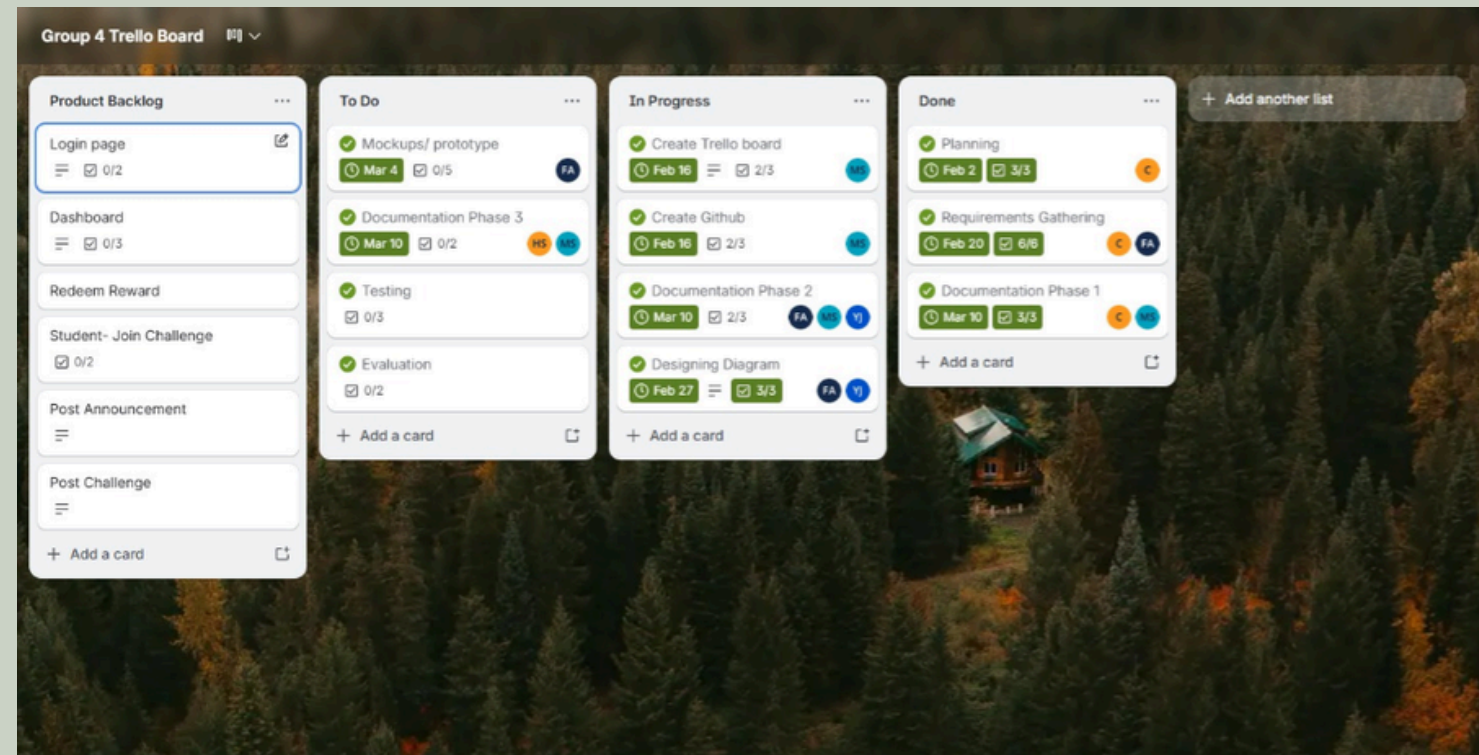


Week 2

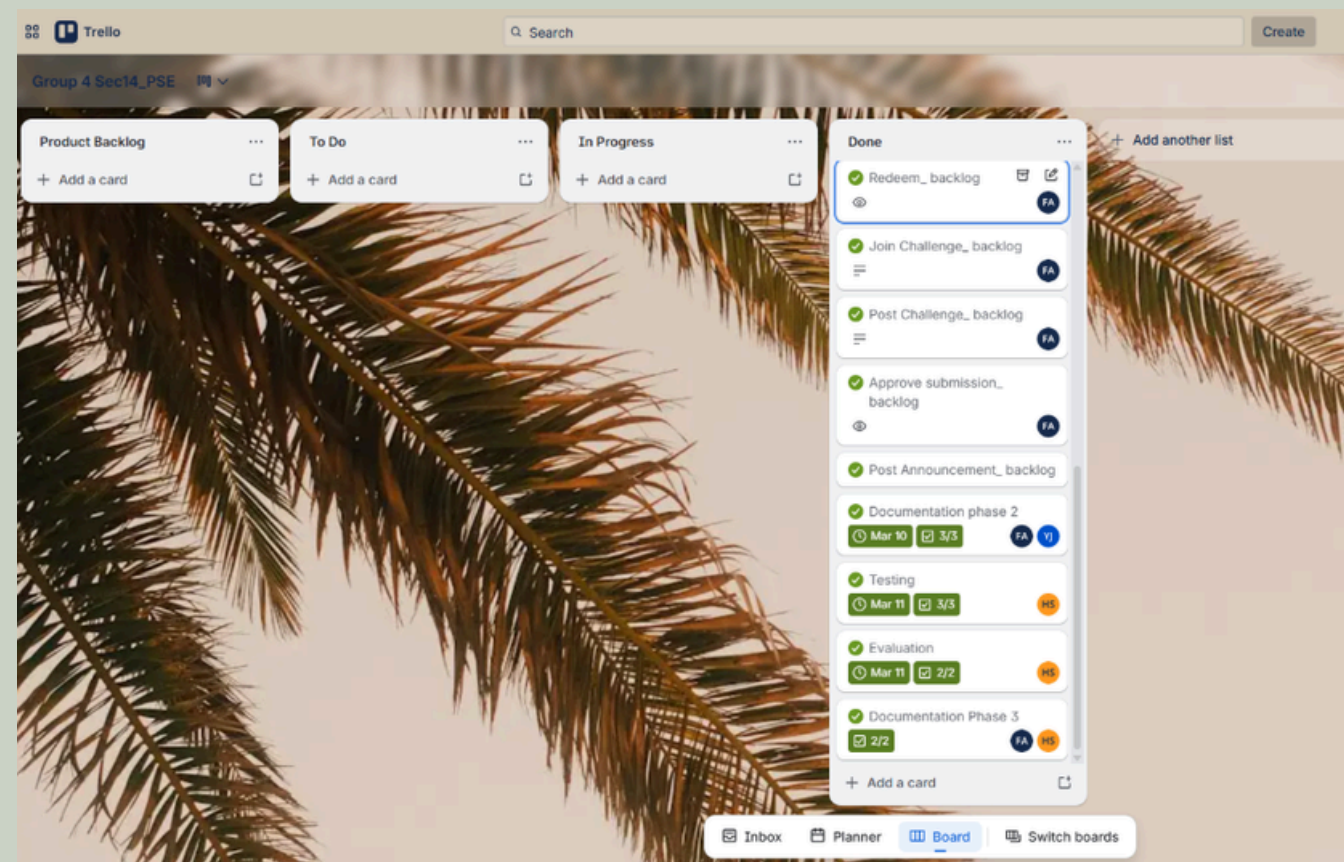


Trello Board

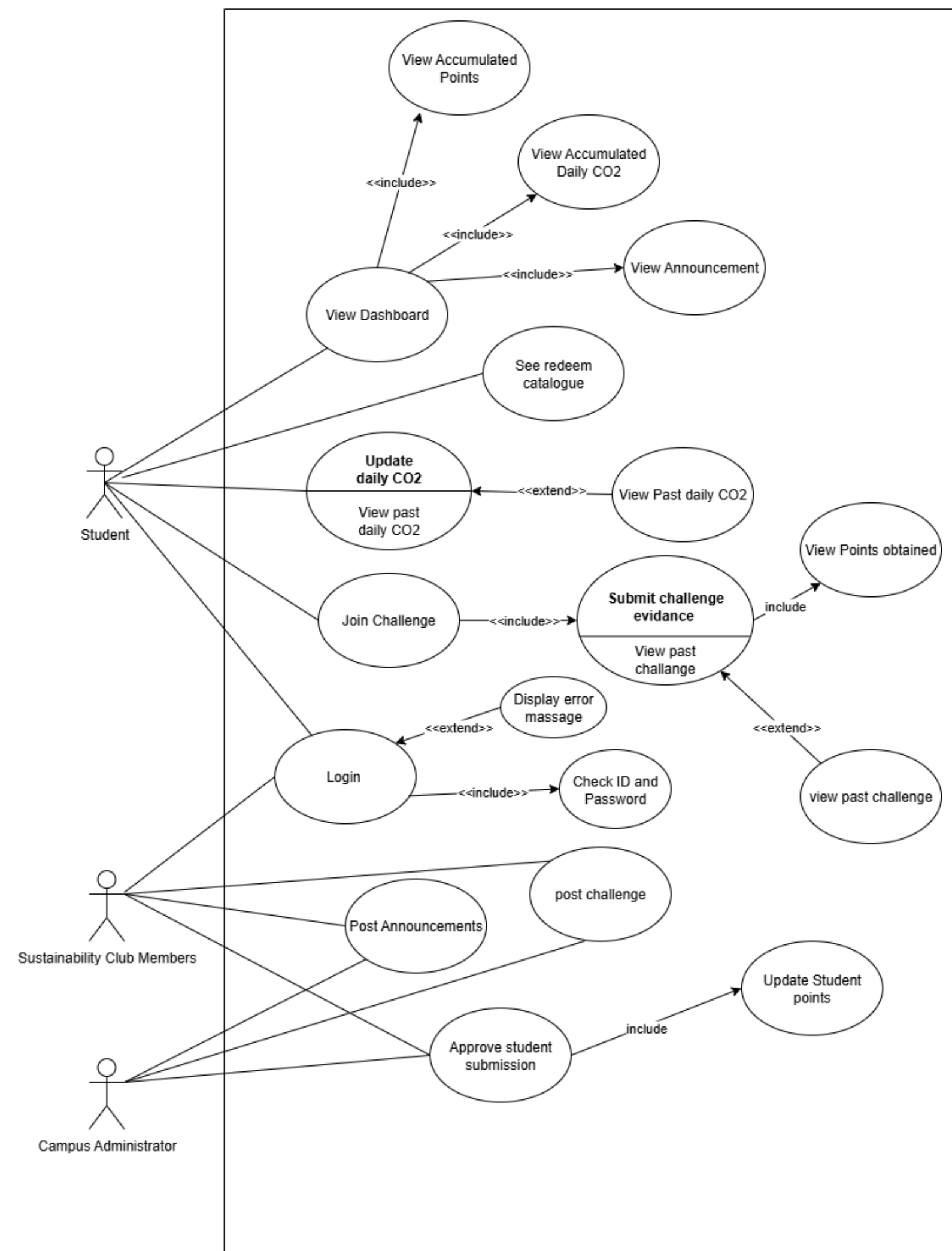
Week 3



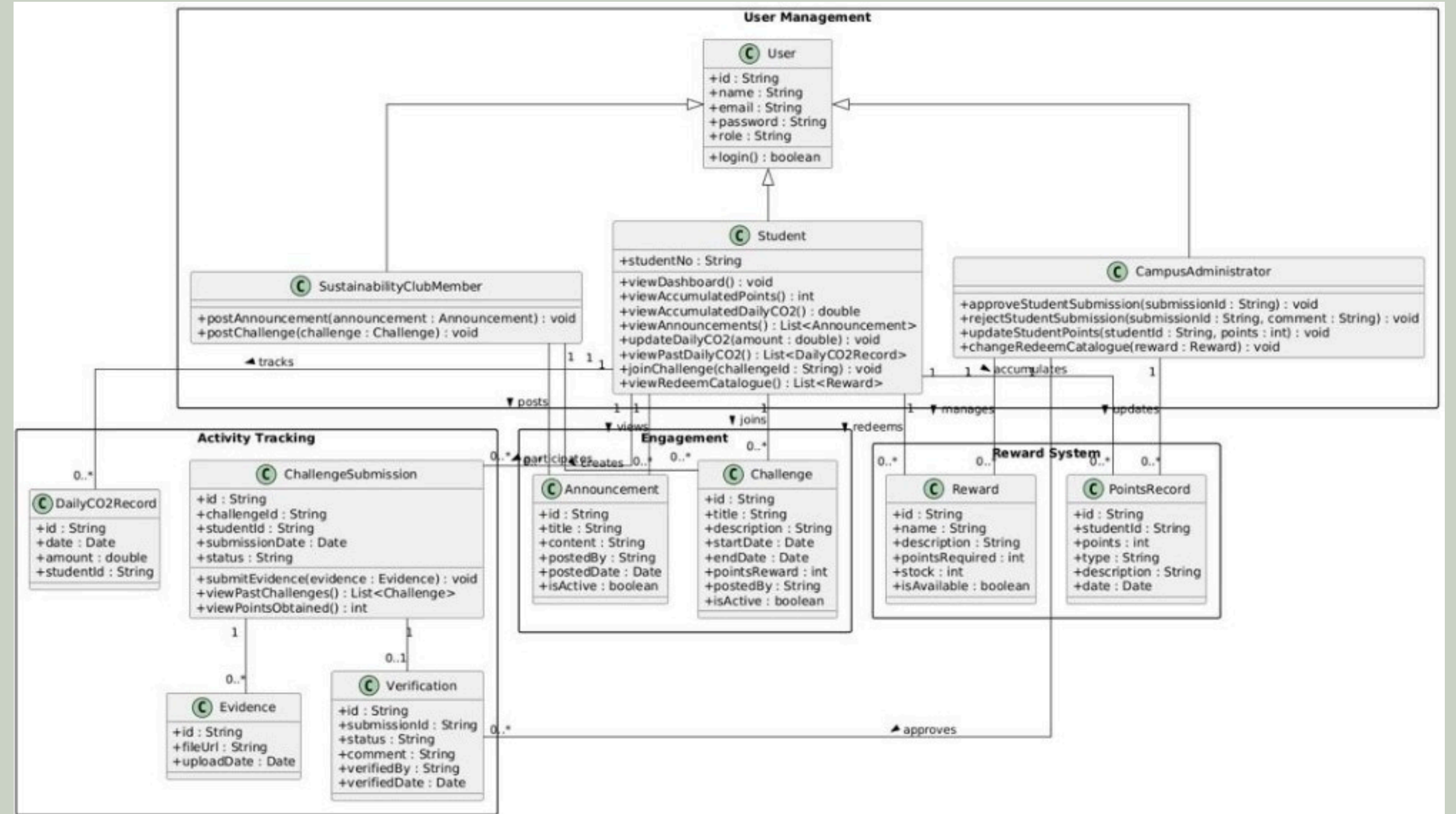
Week 4



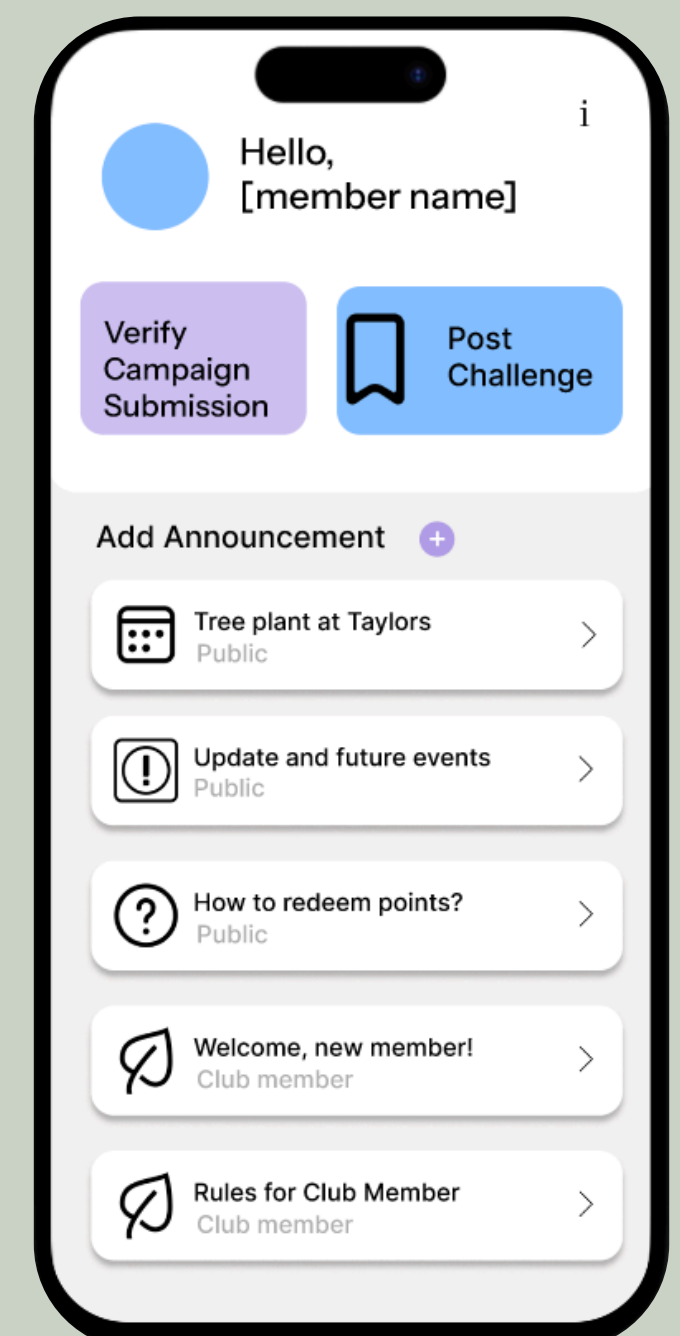
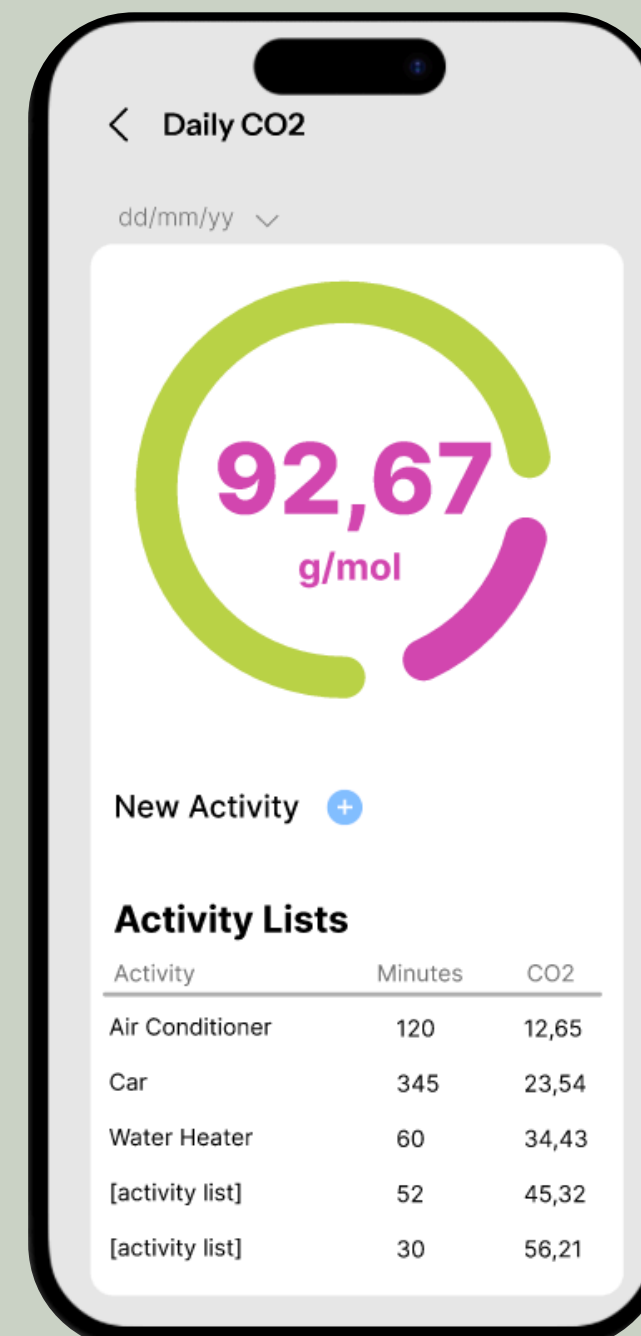
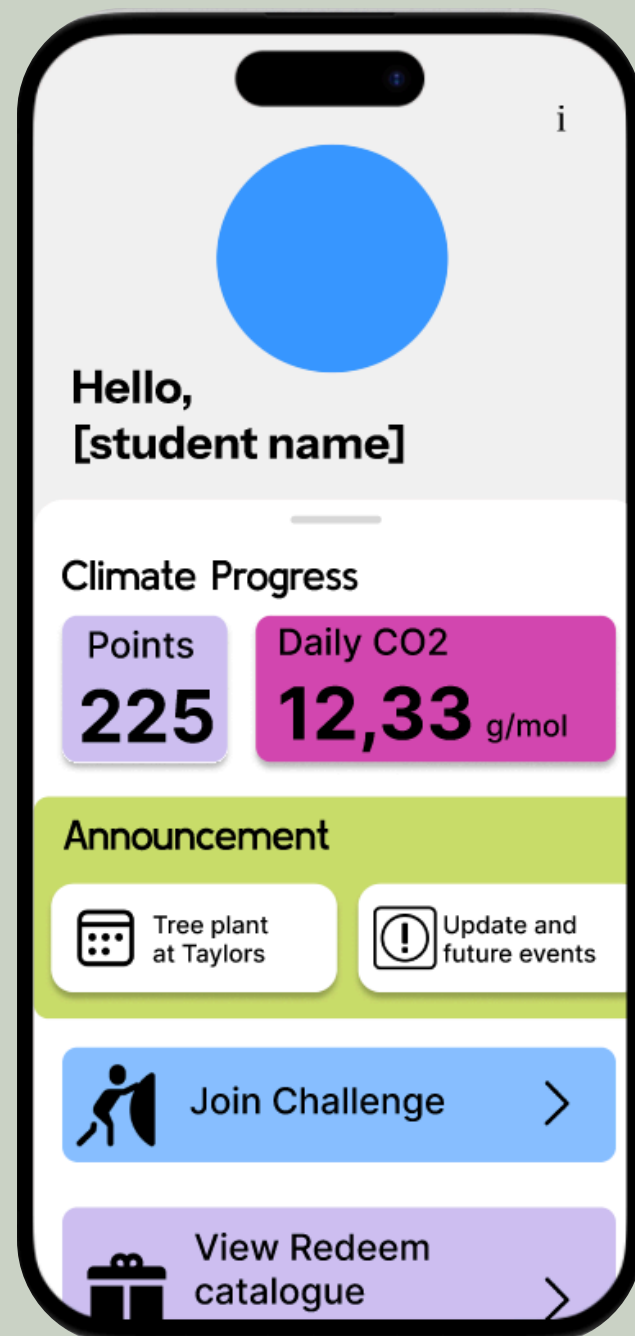
Use Case Diagram



Class Diagram

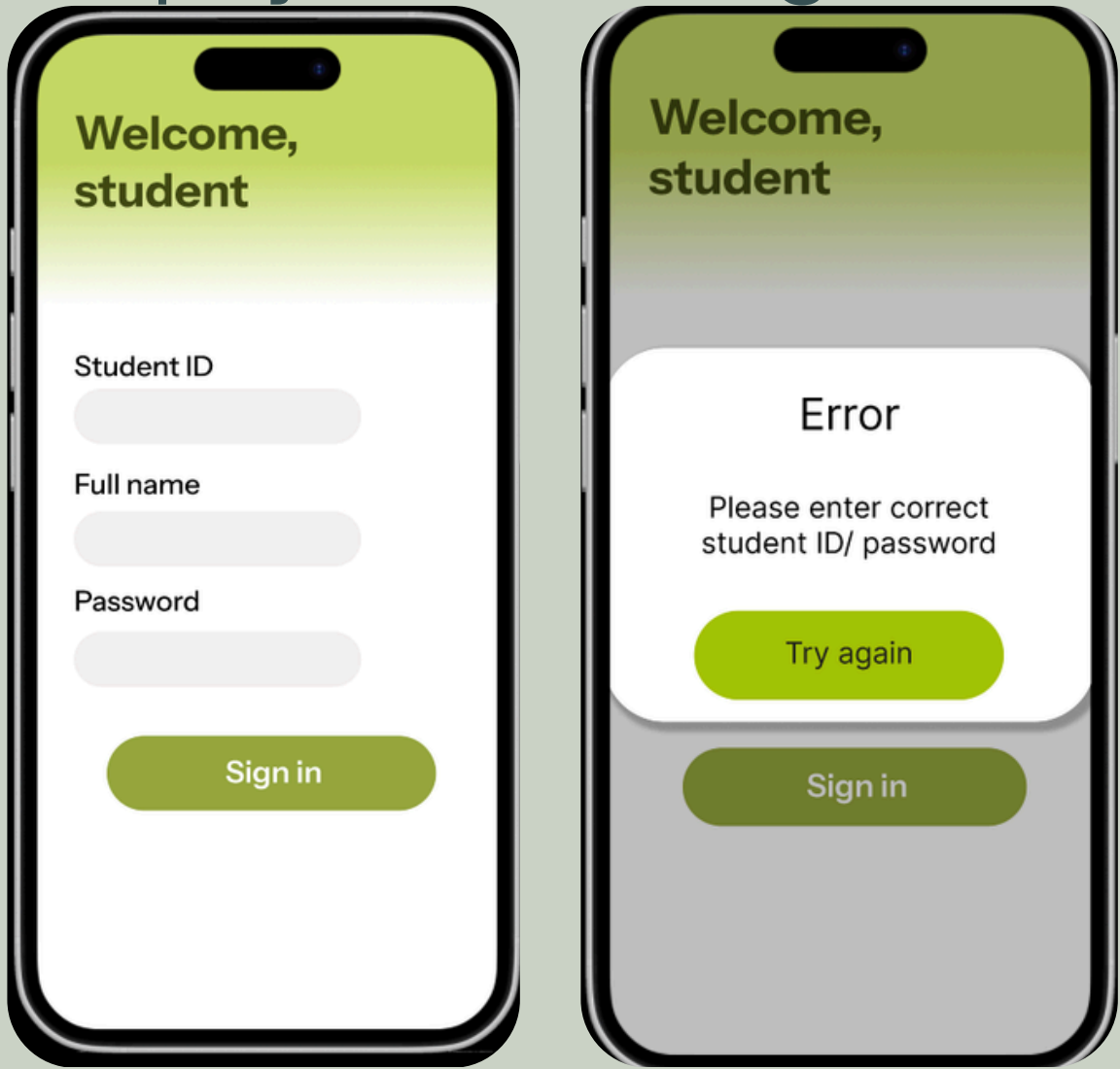


UI Design [Figma Link](#)

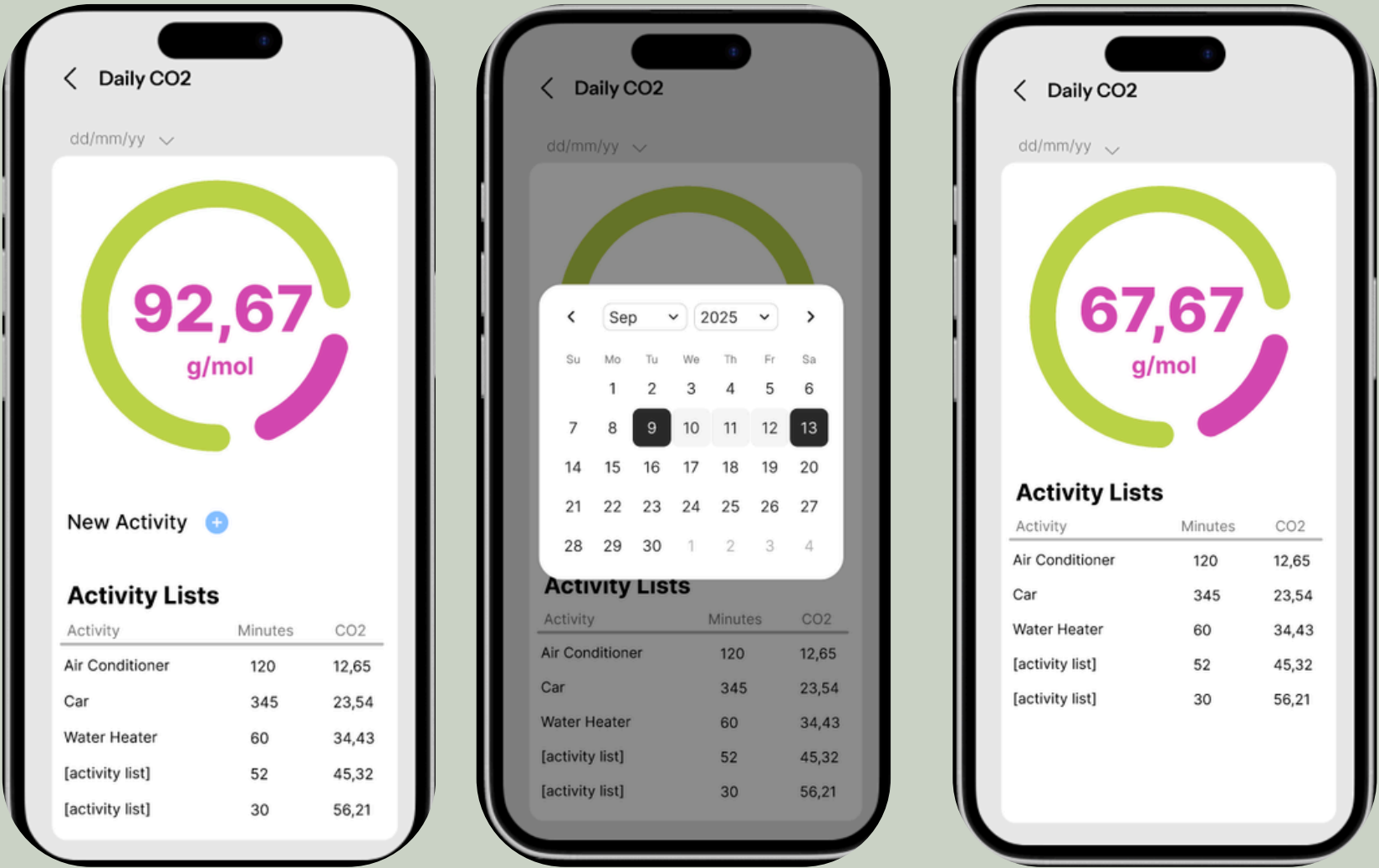


Test Case

Display error massege



View CO2 footprint history



Evaluation

Strengths:

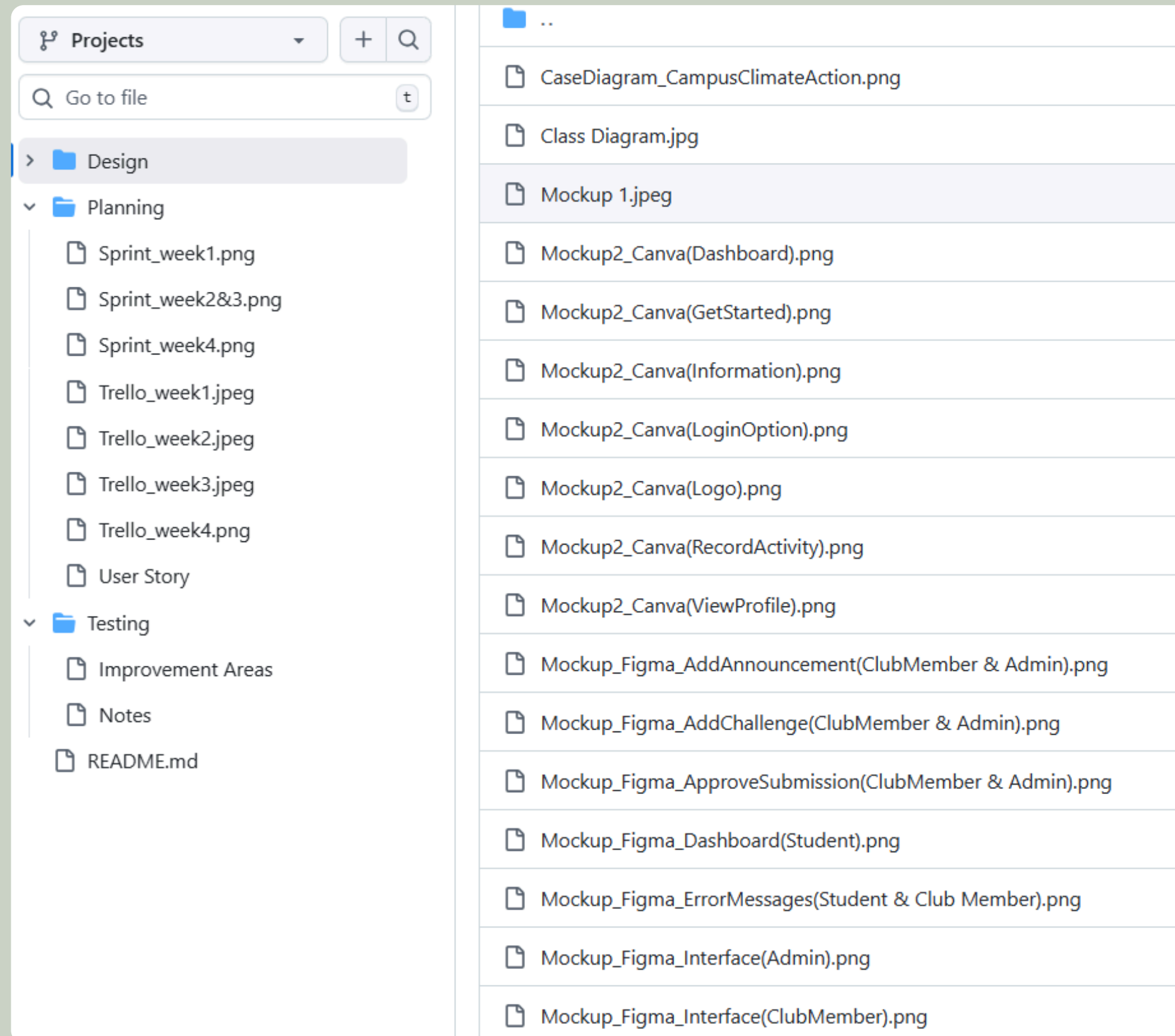
- Simple and clear UI flow is easy to comprehend even for first-time users
- Structured announcement list with consistent layout: title, icon, labels, etc
- Icons and text help improve readability and facilitate scanning of information
- Clean design with minimal distractions to improve ease of navigation

Weaknesses:

- Profile section cannot be edited, limiting personalization
- Students cannot redeem the rewards directly from app, they only can view it
- Announcements can only viewed, not added or interacted with
- No two-step verification, reducing account security

Github Evidence

[Github link](#)



Manage access

☐ Select all

☐  **3486796989-debug**
Awaiting 3486796989-debug's response

☐  **gaosimeng-hub**
Awaiting gaosimeng-hub's response

☐  **seohee-byte**
Awaiting seohee-byte's response

PSE Project

Taylor's University - ITS64604/ITS610204 Group Project
Principles Of Software Engineering

Campus Calimate Action - Climate Action (SDG 13)

Sustainable Development Goal

This app aim for create a lightweight mobile prototype tool adapted to college scenarios, so that students can easily record and calculate the carbon footprint of daily campus life. Motivate students to practice low-carbon behavior and learn more about climate change and set their own goal to achieve low-carbon activity.

Team member

- Gao Simeng - SDG & Planning Leader
- Yao Longjun - System Analyst/ UML Designer
- Felia Alendra - Prototype & UI designer, Agile & Tool Manager
- Hong Seohee - Testing & evaluation lead

Project Description

We proposed Campus Climate Action as the project to fulfill SDG 13- Climate Action. AS a response to climate change we create a software solution, called Campus Climate Action, a campus-initiated application that is intended to assist students in monitoring their carbon footprint and taking part in low-carbon activities. The system enables the students to document the daily activities (commuting, eating, and campus drinking) and subsequently compute the carbon emissions and provide visual reports about their own carbon footprint. It also offers low-carbon challenges, reward systems, and environmental education content to encourage students to engage in permanent contributions to climate-friendly activities.

Reflection and Conclusion

Reflection:

- adopted Agile / Scrum methodology → clear roles (analysis, design, management, testing)
- Tools: Trello (task tracking), Github
- Regular sprint meetings → progress review & task reorganization
- Challenges: limited 4-week timeline, consistency across UML/prototype, incomplete interactivity (announcement/reward buttons)
- Lessons: practical SDLC experience, iterative improvement, teamwork & collaboration enhanced by digital tools
- Future improvements: add interactive feedback, strengthen reward/announcement system, improve accessibility, introduce surveys/gamification

Conclusion:

- Project: Campus Low-Carbon Action App aligned with SDG 13 Climate Action
- Aim: solve campus low-carbon challenges using software engineering + Agile/Scrum
- Process: research → design → prototype → testing → improvement
- Deliverables: UML diagrams, UI sketches, prototype, test plan, functional testing